

# Simulation of mmWave Path Loss under Distance, Frequency, Atmospheric Absorption, Blockage, and Antenna Gain

## Aim

To simulate and analyze the path loss in a millimeter-wave (mmWave) communication system by considering the effects of distance, frequency, atmospheric absorption, blockage, and antenna gain using MATLAB.

## Objectives

1. Study effect of distance.
2. Compare different mmWave frequencies.
3. Include atmospheric absorption and blockage.
4. Observe antenna gain improvement.
5. Plot and analyze path loss.

## Procedure

1. Open MATLAB.
2. Define distance (10–500 m).
3. Select 28, 38 and 60 GHz.
4. Set atmospheric absorption, blockage and antenna gains.
5. Calculate FSPL.
6. Add absorption and blockage losses.
7. Subtract antenna gains.
8. Plot path loss vs distance.
9. Analyze results.

## MATLAB Program

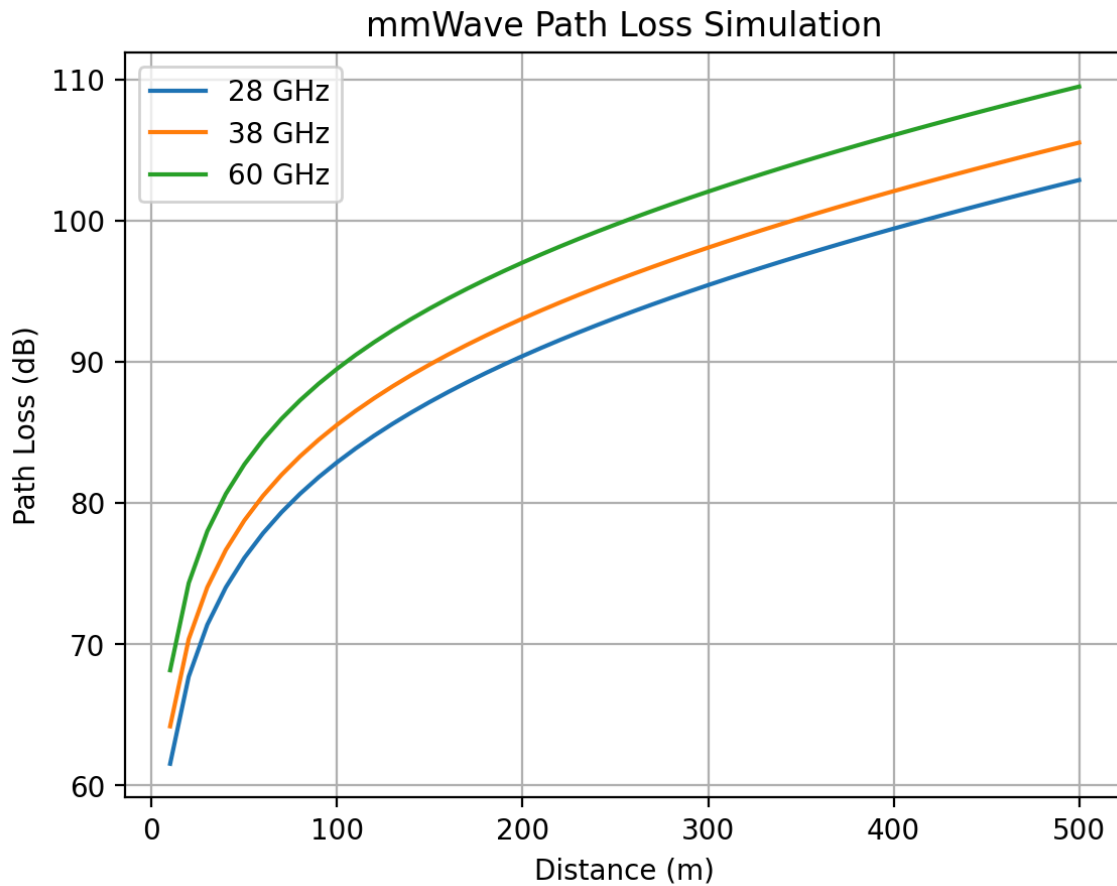
```
clc;
clear;
close all;
d=10:10:500;
f=[28e9 38e9 60e9];
c=3e8;
atm=0.015;
block=20;
Gt=20;
Gr=20;
figure; hold on;
for i=1:length(f)
```

```

FSPL=20*log10(4*pi*d*f(i)/c);
TotalLoss=FSPL+atm*d+block-Gt-Gr;
plot(d,TotalLoss,'LineWidth',2);
end
xlabel('Distance (m)');
ylabel('Path Loss (dB)');
title('mmWave Path Loss Simulation');
legend('28 GHz','38 GHz','60 GHz');
grid on;

```

## Output



A graph of Path Loss (dB) versus Distance (m) is obtained. Path loss increases with distance and frequency. Atmospheric absorption and blockage increase the loss, while antenna gain reduces the effective path loss.

## Result

The simulation confirms that mmWave path loss increases with distance and frequency. High-gain antennas help compensate for atmospheric absorption and blockage losses.